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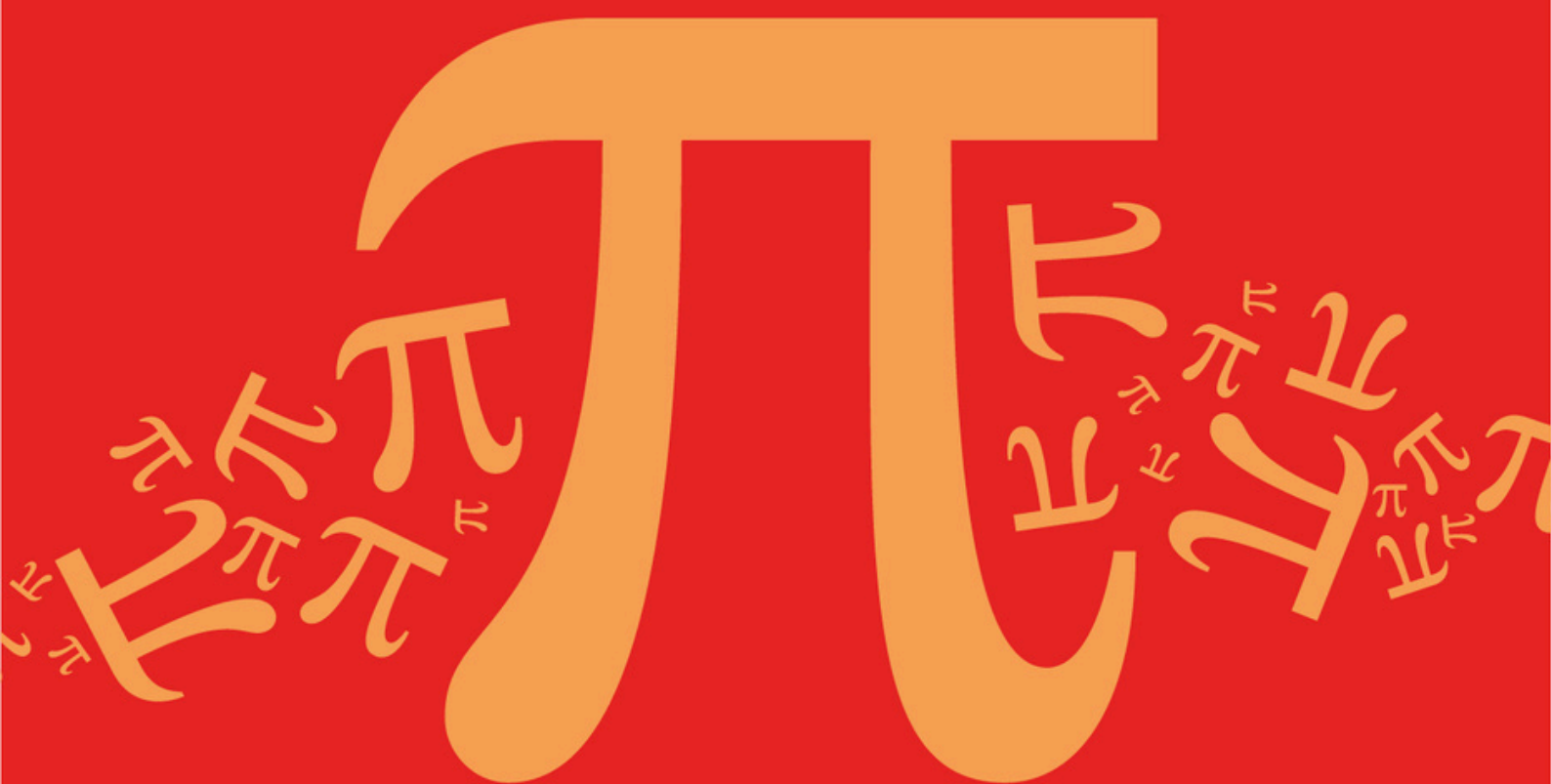
Practice questions created by actual
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Detailed mark scheme

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AQA A Level Further Mathematics (7367)



Topic	B: Complex Numbers
Sub topic	B10: Nth roots

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic

B: Complex Numbers

Sub topic

B10: Nth roots

EXAM PAPERS PRACTICE

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Expresses i or z in polar form	AO1.2	B1	$i = e^{i\frac{\pi}{2}}$
Uses De Moivre's Theorem	AO3.1a	M1	$z = \left[e^{i(\frac{\pi}{2} + 2n\pi)} \right]^{\frac{1}{3}} = \left[e^{i(\frac{\pi}{6} + \frac{2n\pi}{3})} \right]$
Finds three consecutive values for θ	AO1.1a	A1	$\frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$ (or $\frac{3\pi}{2}$ etc) $z = e^{-i\frac{\pi}{2}}, e^{i\frac{\pi}{6}}, e^{i\frac{5\pi}{6}}$
Finds all three correct solutions for z	AO1.1b	A1	ALT $z = e^{i\theta} \Rightarrow z = \cos\theta + i\sin\theta$ $z^3 = i \Rightarrow \cos 3\theta + i\sin 3\theta = i$ $\therefore \cos 3\theta = 0$ and $\sin 3\theta = 1$ $\therefore \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$ (or $\frac{3\pi}{2}$ etc) $z = e^{-i\frac{\pi}{2}}, e^{i\frac{\pi}{6}}, e^{i\frac{5\pi}{6}}$
Total 4 marks			

Q2.

(a) $r = 8$

B1

$\tan^{-1} \pm \frac{4\sqrt{3}}{4}$ or $\pm \frac{\pi}{3}$ seen
 or $\frac{\pi}{6}$ marked as angle to Im axis with
 "vector" in second quadrant on Arg diag

M1

$$\Rightarrow \theta = \frac{2\pi}{3}$$

$$-4 + 4\sqrt{3}i = 8e^{i\frac{2\pi}{3}}$$

A1

3

- (b) (i) modulus of each root = 2

B1✓

use of De Moivre – dividing argument by 3

M1

$$\Rightarrow \theta = \frac{4\pi}{9}, \frac{2\pi}{9}, \frac{8\pi}{9}$$

A1 if 3 “correct” values not all in requested interval

$$2e^{-i\frac{4\pi}{9}}, 2e^{i\frac{2\pi}{9}}, 2e^{i\frac{8\pi}{9}}$$

A2

4

- (ii) Area = $3 \times \frac{1}{2} \times PO \times OR \times \sin \frac{2\pi}{3}$
 Correct expression for area of triangle PQR

M1

$$= 3 \times \frac{1}{2} \times 2 \times 2 \times \sin \frac{2\pi}{3}$$

correct values of lengths in formula

A1

$$= 3\sqrt{3}$$

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3

- (c) Sum of roots (of cubic) = 0
 must be stated explicitly

E1

Sum of 3 roots including Im terms
 in form $r(\cos \theta + i \sin \theta)$

M1

$$2\left(\cos \frac{(-)4\pi}{9} + \cos \frac{2\pi}{9} + \cos \frac{8\pi}{9}\right)$$

isolating real terms; correct and with “2”

A1

$$e^{-i\frac{4\pi}{9}} = \cos \frac{4\pi}{9} - i \sin \frac{4\pi}{9} \text{ seen earlier}$$

$$\text{or } \cos \frac{-4\pi}{9} = \cos \frac{4\pi}{9} \text{ explicitly stated to earn final A1 mark}$$

$$\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} = 0$$

AG

A1cso

4

[14]

Q3.

(a) $1, e^{\frac{2\pi i}{5}}, e^{\frac{4\pi i}{5}}, e^{\frac{-2\pi i}{5}}, e^{\frac{-4\pi i}{5}}$
 accept e^{θ}

B1

1

(b) $\frac{z^5 - 1}{z - 1} = z^4 + z^3 + z^2 + z + 1$
 B0 if assumed

B1

$$= \left(z - e^{\frac{2\pi i}{5}} \right) \left(z - e^{\frac{4\pi i}{5}} \right) \left(z - e^{\frac{-2\pi i}{5}} \right) \left(z - e^{\frac{-4\pi i}{5}} \right)$$

accept if $e^{\frac{6\pi i}{5}}, e^{\frac{8\pi i}{5}}$ used here

M1A1

3

(c) Correct grouping of linear factors

M1

$$e^{\frac{2\pi i}{5}} + e^{\frac{-2\pi i}{5}} = 2\cos \frac{2\pi}{5}$$

clearly shown

A1

$$\left(z^2 - 2\cos \frac{2\pi}{5} z + 1 \right) \left(z^2 - 2\cos \frac{4\pi}{5} z + 1 \right)$$

A1

$\div z^2$ to give answer
 AG

A1

4

(d) Substitute into LHS to give $w^2 + w - 1$

B1

$$\text{RHS } \left(w - 2\cos \frac{2\pi}{5} \right) \left(w - 2\cos \frac{4\pi}{5} \right)$$

B1

$$\text{Solve } w^2 + w - 1 = 0$$

M1

$$w = \frac{-1 \pm \sqrt{5}}{2}$$

A1

$$\cos \frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$$

A1

with reasons for choice

E1

6

[14]

Q4.

(a) (i) $4(1+i\sqrt{3}) = 8\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$
 for either $4(1+i\sqrt{3})$ or $4(1-i\sqrt{3})$ used

M1

$$= 8e^{\frac{j}{3}}$$

If either r or θ is incorrect but the same value in both

(i) and (ii) allow A1 but for θ only if it is given as $\frac{\pi}{6}$

A1

(ii) $4(1-i\sqrt{3}) = 8e^{\frac{-j}{3}}$

A1

3

(b) $z^3 - 4 = \pm\sqrt{-48}$
 taking square root

M1

$$z^3 = 4 \pm 4\sqrt{3}i$$

AG

A1

2

(c) (i) $z = 2e^{\frac{\frac{\pi}{3} + 2k\pi}{3}}$ or $z = 2e^{\frac{\frac{-\pi}{3} + 2k\pi}{3}}$

for the 2; ft incorrect 8, but no decimals

B1F

for either, PI

M1

$$z = 2e^{\frac{\pi}{9}}, 2e^{\frac{7\pi}{9}}, 2e^{\frac{5\pi}{9}}$$

$$2e^{\frac{-\pi}{9}}, 2e^{\frac{-7\pi}{9}}, 2e^{\frac{-5\pi}{9}}$$

Allow A1 for any 2 roots not +/- each other

Allow A2 for any 3 roots not +/- each other

Allow A3 for all 6 correct roots

Deduct A1 for each incorrect root in the interval;

ignore roots outside the interval

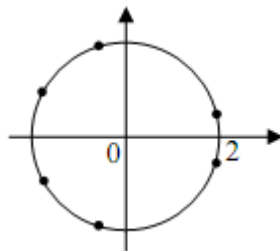
ft incorrect r

A3,2,1F

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5

(ii)



Radius 2

clearly indicated; ft incorrect r

allow B1 for 3 correct points condone lines

B1F

Plotting roots

B2,1

3

(d) (i) Sum of roots = 0 as coefficient of $z^5 = 0$

OE

E1

1

(ii) Use of, say, $\frac{1}{2} \left(e^{\frac{j}{9}} + e^{\frac{-j}{9}} \right) = \cos \frac{\pi}{9}$

M1

$\cos \frac{3\pi}{9} = \frac{1}{2}$ used

A1

$\cos \frac{\pi}{9} + \cos \frac{3\pi}{9} + \cos \frac{5\pi}{9} + \cos \frac{7\pi}{9} = \frac{1}{2}$

AG

A1

3

[17]

Q5.

(a) (i) $\left(e^{\frac{2\pi j}{7}} \right)^7 = e^{2\pi j} = 1$

Or $z^7 = e^{2k\pi j} \quad z = e^{\frac{2k\pi j}{7}} \quad k = 1$

B1

1

(ii) Roots are $\omega^2, \omega^3, \omega^4, \omega^5, \omega^6$

OE; M1A0 for incomplete set

SC B1 for a set of correct roots in terms of $e^{j\theta}$

M1A1

2

(b) Sum of roots considered

M1

= 0

or $\sum_{r=0}^6 \omega^r = \frac{\omega^7 - 1}{\omega - 1} = 0$

A1

2

(c) (i) $\omega^2 + \omega^5 = e^{\frac{4\pi i}{7}} + e^{\frac{10\pi i}{7}}$

M1

$$= e^{\frac{4\pi i}{7}} + e^{\frac{-4\pi i}{7}}$$

Or $\cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} + \cos \frac{4\pi}{7} - i \sin \frac{4\pi}{7}$

A1

$$= 2 \cos \frac{4\pi}{7}$$

AG

A1

3

(ii) $\omega + \omega^6 = 2 \cos \frac{2\pi}{7}; \quad \omega^3 + \omega^4 = 2 \cos \frac{6\pi}{7}$

Allow these marks if seen earlier in the solution

B1,B1

Using part (b)

M1

Result

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A1

4

[12]

Q6.

(a) Correct multiplication of brackets

M1

$$e^{i\theta} + e^{-i\theta} = 2 \cos \theta$$

Clearly shown

A1

2

(b) $2 \cos \theta = 1$

SC If 'hence' not used and, say,
 $z^8 - z^4 + 1 = 0$ is solved by formula, lose

M1

$$\theta = \frac{\pi}{3}$$

M1A1, but then continue M1m1 etc if $\frac{\pi}{3}$ is obtained

A1

$$z^4 = e^{\frac{\pi i}{3}} \text{ or } e^{-\frac{\pi i}{3}}$$

M1

$$z = e^{\frac{\pi i}{12} + \frac{2k\pi i}{4}} \text{ or } e^{-\frac{\pi i}{12} + \frac{2k\pi i}{4}}$$

m1

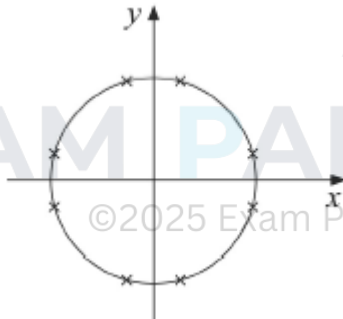
$$e^{\pm \frac{\pi i}{12}}, e^{\pm \frac{7\pi i}{12}}, e^{\pm \frac{5\pi i}{12}}, e^{\pm \frac{11\pi i}{12}}$$

A1 if 3 roots correct

A2,1,0F

6

(c)



B1 for 4 roots indicated correctly on a circle.
CAO

B2,1,0

Indication that $r = 1$

B1

3

[11]

Q7.

(a) $z^4 = 16e^{\frac{4\pi i}{12}}$

Allow M1 if $z^4 = 2e^{\frac{4\pi i}{12}}$

M1

$$= 16 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

OE could be $2ae^{\frac{\pi i}{3}}$ or

$$2a \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

A1

$$= 8 + 8\sqrt{3}i; \alpha = 8$$

ft errors in 2^4

A1F

3

(b) For other roots, $r = 2$

for realising roots are of form $2 \times e^{i\theta}$.

M1 for strictly correct θ

B1

$$\theta = \frac{\pi}{12} + \frac{2k\pi}{4}$$

i.e must be $\left(\text{their } \frac{\pi}{3} + 2k\pi \right) \times \frac{1}{4}$

M1A1

Roots are $2e^{\frac{7\pi i}{12}}, 2e^{\frac{5\pi i}{12}}, 2e^{\frac{-11\pi i}{12}}$

ft error in $\frac{\pi}{12}$ or r

$$\left[\text{accept } 2e^{\left(\frac{\pi}{12} + \frac{2k\pi}{4} \right)i} \quad k = -1, -2, 1 \right]$$

A2,1,0F

5

[8]

Q8.

(a) Any method for finding r or θ

M1

$$r = 4\sqrt{2}, \theta = \frac{\pi}{4}$$

A1A1

(b) $z^5 = 4\sqrt{2} e^{\frac{\pi i}{4}}$

M1 needs some reference to $a + 2k\pi i$

$$z = \sqrt{2} e^{\frac{\pi i}{20} + \frac{2k\pi i}{5}}$$

A1 for r
A1 for θ

incorrect r, θ part (a)

M1A1FA1F

$$z = \sqrt{2} e^{\frac{\pi i}{20}}, \sqrt{2} e^{\frac{9\pi i}{20}}, \sqrt{2} e^{\frac{17\pi i}{20}}, \sqrt{2} e^{\frac{-7\pi i}{20}}, \sqrt{2} e^{\frac{-15\pi i}{20}}$$

Accept r in any form eg $32^{\frac{1}{10}}$
Correct but some answers outside range allow A1
ft incorrect r, θ in part (a)

A2,1,0F

Q9.

(a) $1, e^{\pm \frac{2\pi i}{3}}$

M1 for any method which would lead to the correct answers

Accept e^0 or e^{0i}

Also accept answers written down correctly

M1A1

(b) Any correct method

M1

Shown for one root

AG

A1

(c) (i) $\frac{\omega}{\omega + 1} = \frac{\omega}{-\omega^2}$

ie use of result in (b)

$$= -\frac{1}{\omega}$$

AG

A1

2

(ii)
$$\frac{\omega^2}{\omega^2 + 1} = -\omega$$

AG

A1

1

(iii)
$$\left(\frac{\omega}{\omega + 1}\right)^k + \left(\frac{\omega^2}{\omega^2 + 1}\right)^k = \left(-\frac{1}{\omega}\right)^k + (-\omega)^k$$

M1A1

Use of $\omega = e^{\frac{2\pi i}{3}}$

m1

$$= (-1)^k \left(e^{-\frac{2k\pi i}{3}} + e^{\frac{2k\pi i}{3}} \right)$$

A1

$$= (-1)^k 2 \cos \frac{2k\pi}{3}$$

AG

A1

5

[12]

Q10.

(a) (i)
$$z^3 = \frac{4 \pm \sqrt{16 - 32}}{2}$$

M1

$$= 2 \pm 2i$$

AG

A1

2

(ii)
$$2 + 2i = 2\sqrt{2}e^{\frac{\pi i}{4}}, \quad 2 - 2i = 2\sqrt{2}e^{-\frac{\pi i}{4}}$$

M1 for either result or for one of $r = 2\sqrt{2}$, $\theta = \pm \frac{\pi}{4}$

M1

$$\left(r = 2\sqrt{2} \text{ A1, } \theta = \pm \frac{\pi}{4} \text{ A1} \right)$$

A1A1

$$z = \sqrt{2}e^{\frac{\pi i}{12} + \frac{2\pi i}{3}} \text{ or } \sqrt{2}e^{\frac{-\pi i}{12} + \frac{2\pi i}{3}}$$

M1 for either

M1

$$z = \sqrt{2}e^{\frac{\pm \pi i}{12}}, \sqrt{2}e^{\frac{\pm 3\pi i}{4}}, \sqrt{2}e^{\frac{\pm 7\pi i}{12}}$$

allow A1 for any 3 correct ft errors in $\pm \frac{\pi}{4}$

A2,1,0 F

6

(b) Multiplication of brackets

M1

Use of $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$
AG

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A1

2

(c)
$$\left(z - \sqrt{2}e^{\frac{\pi i}{12}} \right) \left(z - \sqrt{2}e^{-\frac{\pi i}{12}} \right)$$

$$= z^2 - 2\sqrt{2} \cos \frac{\pi}{12} z + 2$$

PI

M1A1F

$$\left(z^2 - 2\sqrt{2} \cos \frac{\pi}{12} z + 2 \right)$$

Product is $\left(z^2 - 2\sqrt{2} \cos \frac{7\pi}{12} z + 2 \right)$

(or $z^2 + 2z + 2$)

A1F

3

$$(z^2 - 2\sqrt{2} \cos \frac{3\pi}{4} z + 2)$$

[13]

Q11.

(a) $z = e^{\frac{2k\pi i}{6}}, \quad k = 0, \pm 1, \pm 2, 3$

M1

OE

M1A1 only if:

(1) range for k is incorrect e.g. $0, 1, 2, 3, 4, 5$

(2) i is missing

A2,1,0

3

(b) (i) $\frac{w^2 - 1}{w} = w - \frac{1}{w} = 2i \sin \theta$

AG

M1A1

2

(ii) $\frac{w}{w^2 - 1} = \frac{1}{2i \sin \theta}$

M1

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$$= -\frac{i}{2 \sin \theta}$$

AG

A1

2

(iii) $\frac{2i}{w^2 - 1} = \frac{-2iw^{-1}i}{2 \sin \theta}$

Or for $\frac{1}{\sin \theta e^{i\theta}}$

M1

$$= \frac{1}{\sin \theta} (\cos \theta - i \sin \theta)$$

A1

$$= \cot \theta - i$$

AG

A1

3

(iv) $z = \frac{2i}{w^2 - 1}$ Or $z + 2i = \frac{2i}{w^2 - 1} + 2i$
ie any correct method

M1

$$z + 2i = zw^2$$

AG

A1

2

(c) (i) No coefficient of z^6

E1

1

(ii) $(w^2)^6 = 1$ $w^2 = e^{\frac{k\pi i}{3}}$

B1

$$z = \cot \frac{k\pi}{6} - i, \quad k = \pm 1, \pm 2, 3$$

M1

A2, 1, 0

Alternatively:

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$$z + 2i = e^{\frac{k\pi i}{3}} z$$

B1

$$z = \frac{2i}{e^{\frac{k\pi i}{3}} - 1}$$

M1

roots A2, 1, 0

(NB roots are $\pm \sqrt{3} - i; \pm \frac{1}{\sqrt{3}} - i; -i$)

4

[17]

Examiner reports

Q2.

- (a) Many students seemed unable to find the correct argument of the complex number. Students would be well advised to make a small sketch of the Argand diagram in order to determine the correct quadrant for the angle θ .
- (b) (i) An incorrect value of θ in part (a) prevented a large number of students from obtaining the correct roots of the cubic equation. Most students had the correct modulus and realised the need to divide an argument by 3, but did not always apply de Moivre's Theorem correctly.
- (ii) It was possible for students to score full marks here if they had the correct modulus of the 3 roots, provided they recognised that the points P , Q and R were the vertices of an equilateral triangle. Poor trigonometrical skills prevented most students from obtaining the correct value for the area of the triangle.
- (c) It was necessary to state explicitly that the sum of the roots of the cubic equation was zero. By considering the sum of the roots and then taking real parts of both sides of the equation the printed result followed. Better students stated explicitly that $\cos\left(-\frac{4\pi}{9}\right) = \cos\left(\frac{4\pi}{9}\right)$ at some stage in their proof.

Q3.

Responses to this question were extremely poor and very few candidates were able to make any headway at all after part (a). Very few candidates seemed to understand what linear factors were – most merely wrote down the roots of z^5 in some other form. This in turn meant that virtually no one was able to answer part (c). There were not many serious attempts at part (d) in spite of the fact that it could be done by assuming the result in part (c), although some candidates were able to pick up the odd mark.

Q4.

Parts (a) and (b) of this question were well done; part (c)(i) was reasonably well done especially by the more able candidates. However, the Argand diagrams were very poorly drawn. Many candidates drew a rough circle by hand, with no indication of its radius, and then went on to put crosses on their circle at points which bore little resemblance to the angles in question. Some candidates drew no circle at all but drew lines of differing lengths from the origin. There were few correct responses to part (d) as candidates failed to realise the relevance of part (c), and especially part (d)(i), to this last part.

Q5.

Whilst part (a) was generally well done, relatively few candidates expressed the other roots in terms of ω , but rather gave them in the form $re^{i\theta}$.

Few, also, were able to complete part (b). The relation $1 + \omega + \omega^2 = 0$ appeared with regularity.

Part (c) was poorly answered. Although correct answers were written down as they were given in the question, few responses were convincing and as has already been stated

earlier, if answers are given, it is the responsibility of the candidates to supply sufficient working to convince the examiner that they understand the methods involved.

Q6.

Part (a) was quite well done, although for some reason the multiplication of the brackets sometimes resulted in $z^8 - 2(e^{i\theta} + e^{-i\theta})z^4 + 1$ followed by the printed result. In part (b), a number of candidates lost some marks through attempting to use an 'otherwise' method

instead of the 'hence' method as directed. Those candidates arriving at $z^4 = e^{\frac{\pi i}{3}}$ usually went on to solve the given equation correctly. The Argand diagram, however, was poorly drawn. Frequently no circle was indicated and roots appeared at different distances from the origin, and in many cases candidates seemed to think that the eight roots were equally spaced round the origin.

Q7.

This question proved to be slightly more demanding for candidates than had been anticipated. The main difficulty in part (a) was that, having written down $2^4 e^{\frac{\pi i}{3}}$ (or

frequently incorrectly $2e^{\frac{\pi i}{3}}$), candidates were unsure about how to proceed, and they either abandoned this part of the question at that point or then tried to manipulate $a(1 + \sqrt{3}i)$ with little success.

In part (b), candidates were divided into two clear categories: those who had been clearly taught how to find roots of equations in the form $re^{i\theta}$ and those who were very muddled in their thinking. It generally hinged on candidates' ability to write θ as $\frac{\pi}{12} + \frac{k\pi}{2}$. Quite a number of otherwise correct solutions lost a mark through giving values of θ outside the specified range.

Q8.

Part (a) was done well by the majority of candidates. However, responses to part (b) were less successful. A number of candidates gave the roots of $z^5 = 1$ as their answer to part (b), whilst others left the modulus of the roots as $4\sqrt{2}$ instead of $\sqrt{2}$, and others again gave solutions outside the range of θ as specified in the question. Some candidates yet

again were either unable to handle $\frac{1}{5} \left(\frac{\pi}{4} + 2k\pi \right) i$ or, when taking the fifth root of $e^{\left(\frac{\pi}{4} + 2k\pi \right) i}$, wrote $e^{\left(\frac{\pi}{4} + \frac{2k\pi}{5} \right) i}$.

Q9.

It was disappointing to find many candidates unsure of the cube roots of unity and even more unsure of how to obtain them. It was also disappointing to note that few candidates were able to establish the result $1 + \omega + \omega^2 = 0$ in part (b), in spite of the variety of ways in which this result could be established. On the whole, parts (c)(i) and (c)(ii) were correctly done in spite of using roundabout methods to obtain the printed results. In part (c)(iii),

however, most solutions ended at $\left(-\frac{1}{\omega}\right)^k + (-\omega)^k$ but of those candidates who attempted this part further, sign errors hindered completely correct solutions.

Q10.

Candidates were usually able to establish the result in part (a) although the methods used were sometimes somewhat inelegant. Part (a)(ii) was reasonably well done although some carelessness was in evidence in this part. For instance, some candidates although

showing that the argument of z^3 was $\pm \frac{1}{4}\pi$ continued their solution with only $+\frac{1}{4}\pi$ and so arrived at a total of three roots. Others having reached $|z^3| = \sqrt{8}$ then thought that $|z| = \sqrt{8}$ also.

A few candidates used a method which, although possible, was not really suitable. They replaced the z^3 in $z^6 - 4z^3 + 8 = 0$ with $2 \pm 2i$ and so arrived at $z^6 = \pm 8i$. This latter equation gave the twelve roots of $z^{12} = -64$ and the method was incomplete unless 6 of the roots were rejected. Part (b) was generally well done, but part (c) was really only completed by candidates who had correctly answered part (a)(ii).

Q11.

Although part (a) of this question was standard work it was surprising to see many candidates fail to obtain full marks. The commonest errors were either to express the six roots of $z^6 = 1$ in the form $a + bi$, or to give the roots in the range 0 to $27i$. A few candidates wrote down the 6 roots as $e^{\frac{k\pi i}{3}}$ with $k = \pm 1, \pm 2, \pm 3$. In part (b), parts (i) and (iv) were often well done, but relatively few candidates spotted w part (b)(ii) as the reciprocal of part

(b)(i), and it was not unusual to see $\frac{w}{w^2 - 1}$ rewritten as $w^{-1} - w$.

Part (b)(iii) was beyond all but the most able candidates although quite a number arrived

at $\frac{1}{\sin \theta e^{i\theta}}$ at which point their solutions usually petered out. There was a wide variety of reasons why the equation $(z + 2i) = z^6$ had only 5 roots with about 50% of them spurious. In part (c)(ii) only one or two candidates used the hints given in the earlier parts of the question, but instead, solved the equation $(z + 2i)^6 = z^6$ from first principles by writing $z + 2i$

$$= ze^{\frac{k\pi i}{3}} \text{ followed by } z = \frac{2i}{e^{\frac{k\pi i}{3}} - 1}.$$

Of the few serious attempts made by candidates at this part of the question, most solutions ended at the point indicated and only the most able candidates found the five roots of the equation in the required form.